

Table of contents

\\USER

<table border="1"><tr><td>Polimeni</td></tr></table>	Polimeni
Polimeni	

<table border="1"><tr><td><table border="1"><tr><td>sebp</td></tr></table></td></tr></table>	<table border="1"><tr><td>sebp</td></tr></table>	sebp
<table border="1"><tr><td>sebp</td></tr></table>	sebp	
sebp		

<table border="1"><tr><td><table border="1"><tr><td>multiVENC_sub01</td></tr></table></td></tr></table>	<table border="1"><tr><td>multiVENC_sub01</td></tr></table>	multiVENC_sub01
<table border="1"><tr><td>multiVENC_sub01</td></tr></table>	multiVENC_sub01	
multiVENC_sub01		

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>BEAT_FQ_venc_4_6_8_13_40</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>BEAT_FQ_venc_4_6_8_13_40</td></tr></table></td></tr></table>	<table border="1"><tr><td>BEAT_FQ_venc_4_6_8_13_40</td></tr></table>	BEAT_FQ_venc_4_6_8_13_40
<table border="1"><tr><td><table border="1"><tr><td>BEAT_FQ_venc_4_6_8_13_40</td></tr></table></td></tr></table>	<table border="1"><tr><td>BEAT_FQ_venc_4_6_8_13_40</td></tr></table>	BEAT_FQ_venc_4_6_8_13_40	
<table border="1"><tr><td>BEAT_FQ_venc_4_6_8_13_40</td></tr></table>	BEAT_FQ_venc_4_6_8_13_40		
BEAT_FQ_venc_4_6_8_13_40			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>BEAT_FQ_venc_4_5_7_10_20</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>BEAT_FQ_venc_4_5_7_10_20</td></tr></table></td></tr></table>	<table border="1"><tr><td>BEAT_FQ_venc_4_5_7_10_20</td></tr></table>	BEAT_FQ_venc_4_5_7_10_20
<table border="1"><tr><td><table border="1"><tr><td>BEAT_FQ_venc_4_5_7_10_20</td></tr></table></td></tr></table>	<table border="1"><tr><td>BEAT_FQ_venc_4_5_7_10_20</td></tr></table>	BEAT_FQ_venc_4_5_7_10_20	
<table border="1"><tr><td>BEAT_FQ_venc_4_5_7_10_20</td></tr></table>	BEAT_FQ_venc_4_5_7_10_20		
BEAT_FQ_venc_4_5_7_10_20			

<table border="1"><tr><td><table border="1"><tr><td><table border="1"><tr><td>ToF</td></tr></table></td></tr></table></td></tr></table>	<table border="1"><tr><td><table border="1"><tr><td>ToF</td></tr></table></td></tr></table>	<table border="1"><tr><td>ToF</td></tr></table>	ToF
<table border="1"><tr><td><table border="1"><tr><td>ToF</td></tr></table></td></tr></table>	<table border="1"><tr><td>ToF</td></tr></table>	ToF	
<table border="1"><tr><td>ToF</td></tr></table>	ToF		
ToF			

\\USER\Polimeni\sebp\multiVENC_sub01\BEAT_FQ_venc_4_6_8_13_40

TA: 4:29 min Coil Selection: Manual Voxel Size: 0.5×0.5×2.2 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.0 A2.5 H23.7 mm
Orientation	C > T42.2
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	2.2 mm
TR	62.22 ms
TE	7.47 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7

Contrast - Common

TR	62.22 ms
TE	7.47 ms
Flip Angle Mode	Constant
Flip Angle	50 deg
Dark Blood	Off
Wrap-up Magn.	None
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	24
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s
Pause after Meas. 4	0.0 s
Pause after Meas. 5	0.0 s
Pause after Meas. 6	0.0 s

Contrast - Dynamic

Pause after Meas. 7	0.0 s
Pause after Meas. 8	0.0 s
Pause after Meas. 9	0.0 s
Pause after Meas. 10	0.0 s
Pause after Meas. 11	0.0 s
Pause after Meas. 12	0.0 s
Pause after Meas. 13	0.0 s
Pause after Meas. 14	0.0 s
Pause after Meas. 15	0.0 s
Pause after Meas. 16	0.0 s
Pause after Meas. 17	0.0 s
Pause after Meas. 18	0.0 s
Pause after Meas. 19	0.0 s
Pause after Meas. 20	0.0 s
Pause after Meas. 21	0.0 s
Pause after Meas. 22	0.0 s
Pause after Meas. 23	0.0 s
Multiple Series	Off
Reordering	Linear

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	2.2 mm
Base Resolution	320
Phase Resolution	56 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.0 A2.5 H23.7 mm
Orientation	C > T42.2
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %

Geometry - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	2.2 mm
TR	62.22 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.0 A2.5 H23.7 mm
Orientation	C > T42.2
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	R2.0 A2.5 H23.7
Phase	-2.0 mm
Read	-15.9 mm
Shift	-17.8 mm
Initial Orientation	C > T
C > T	42.20
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	IsoCenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.254010 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	62.22 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	160 mm
FOV Phase	100.0 %
Phase Resolution	56 %
Cine	On
Trajectory	Cartesian
Dummy Heartbeats	0
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Flow

Flow Mode	Single Dir.
Encodings	5
Velocity Enc. 1	4 cm/s
Velocity Enc. 2	6 cm/s
Velocity Enc. 3	8 cm/s
Velocity Enc. 4	13 cm/s
Velocity Enc. 5	40 cm/s
Direction	Through Plane
Rephased Images	On
Magnitude Images	On
Magnitude Sum	Off
Phase Images	On

Inline - Subtraction

Subtract	Off
Measurements	24
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl_r
Dimension	2D
Sequence Type	Gre
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Performance*
Flow Compensation	On
Reordering	Linear
Bandwidth	391 Hz/Px
Echo Spacing	10.37 ms
Asymmetric Echo	Off
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE TR

\\USER\Polimeni\sebp\multiVENC_sub01\BEAT_FQ_venc_4_5_7_10_20

TA: 4:29 min Coil Selection: Manual Voxel Size: 0.5×0.5×2.2 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Pause after Meas. 7	0.0 s
Pause after Meas. 8	0.0 s
Pause after Meas. 9	0.0 s
Pause after Meas. 10	0.0 s
Pause after Meas. 11	0.0 s
Pause after Meas. 12	0.0 s
Pause after Meas. 13	0.0 s
Pause after Meas. 14	0.0 s
Pause after Meas. 15	0.0 s
Pause after Meas. 16	0.0 s
Pause after Meas. 17	0.0 s
Pause after Meas. 18	0.0 s
Pause after Meas. 19	0.0 s
Pause after Meas. 20	0.0 s
Pause after Meas. 21	0.0 s
Pause after Meas. 22	0.0 s
Pause after Meas. 23	0.0 s
Multiple Series	Off
Reordering	Linear

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.0 A2.5 H23.7 mm
Orientation	C > T42.2
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	2.2 mm
TR	62.22 ms
TE	7.47 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	HC1-7

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	2.2 mm
Base Resolution	320
Phase Resolution	56 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Off

Contrast - Common

TR	62.22 ms
TE	7.47 ms
Flip Angle Mode	Constant
Flip Angle	50 deg
Dark Blood	Off
Wrap-up Magn.	None
Reconstruction	Magn./Phase

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	24
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s
Pause after Meas. 4	0.0 s
Pause after Meas. 5	0.0 s
Pause after Meas. 6	0.0 s

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	R2.0 A2.5 H23.7 mm
Orientation	C > T42.2
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %

Geometry - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	2.2 mm
TR	62.22 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	R2.0 A2.5 H23.7 mm
Orientation	C > T42.2
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	R2.0 A2.5 H23.7
Phase	-2.0 mm
Read	-15.9 mm
Shift	-17.8 mm
Initial Orientation	C > T
C > T	42.20
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	IsoCenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.254010 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	62.22 ms
Segments	1
Concatenations	1

Physio - Cardiac

Dark Blood	Off
FOV Read	160 mm
FOV Phase	100.0 %
Phase Resolution	56 %
Cine	On
Trajectory	Cartesian
Dummy Heartbeats	0
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Flow

Flow Mode	Single Dir.
Encodings	5
Velocity Enc. 1	4 cm/s
Velocity Enc. 2	5 cm/s
Velocity Enc. 3	7 cm/s
Velocity Enc. 4	10 cm/s
Velocity Enc. 5	20 cm/s
Direction	Through Plane
Rephased Images	On
Magnitude Images	On
Magnitude Sum	Off
Phase Images	On

Inline - Subtraction

Subtract	Off
Measurements	24
StdDev	Off
Motion Correction	None
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
------------------	-----

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl_r
Dimension	2D
Sequence Type	Gre
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Performance*
Flow Compensation	On
Reordering	Linear
Bandwidth	391 Hz/Px
Echo Spacing	10.37 ms
Asymmetric Echo	Off
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE TR

\\USER\Polimeni\sebp\multiVENC_sub01\ToF

TA: 20 sec Coil Selection: Auto Voxel Size: 2.3×2.3×2.3 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R2.0 P27.9 H23.2 mm
Orientation	C > T1.4
Phase Encoding Dir.	R >> L
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	87.5 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	2.34 mm
TR	6.3 ms
TE	2.95 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	6.3 ms
TE	2.95 ms
MTC	Off
Magn. Preparation	None
Flip Angle	18 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	2.34 mm
Base Resolution	128
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	R2.0 P27.9 H23.2 mm
Orientation	C > T1.4
Phase Encoding Dir.	R >> L
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	87.5 %
FOV Read	300 mm
FOV Phase	100.0 %
Slice Thickness	2.34 mm
TR	6.3 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	R2.0 P27.9 H23.2 mm
Orientation	C > T1.4
Phase Encoding Dir.	R >> L

Geometry - AutoAlign

AutoAlign	---
Initial Position	R2.0 P27.9 H23.2
Phase	-2.0 mm
Read	-23.8 mm
Shift	27.4 mm
Initial Orientation	C > T
C > T	1.40
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
CoilShim	Off
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.254010 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6.3 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	300 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
------------------	-----

Inline - MapIt

MapIt	None
Flip Angle	18 deg
Measurements	1
Contrasts	1
TE	2.95 ms
TR	6.3 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Fast
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s